QUIZ-1

1- Write a program that computes the results of function F(x) for x between 1 and 100

$$F(x) = 3{,}4\, e^3 + 18{,}7x - 1{,}6x^2 + x^4 Sin\,(30^0) + \ln\left(\frac{x}{2{,}5}\right)$$

and prints the results as

x	F(x)
1	…
2	…
…	…

2- Write a program that computes, using the **FDA**, **CDA**, **CDA2** at x=4.7, with h=0.01, the first and second derivative of the function:

$$F(x) = 3.4 + 18.7x - 1.6x^2$$

$$\boldsymbol{FDA} = \frac{\big(f(x+h) - f(x)\big)}{h}$$

$$\boldsymbol{CDA} = \frac{\big(f(x+h) - f(x-h)\big)}{2h}$$

$$\boldsymbol{CDA2} = \frac{\big(f(x-h) - 2f(x) + f(x+h)\big)}{h^2}$$

FDA = (f(x+h)-f(x))/h / **CDA** = (f(x+h)-f(x-h))/(2h) / **REA2** = (-f(x-2h)+16f(x-h)-30f(x)+16f(x+h)-f(x+2h))/(12h^2)

3. A quadratic equation is represented in the form $\boldsymbol{ax^2 + bx + c = 0}$. The discriminant ($\Delta$) and the roots ($x_1, x_2$) of this equation are calculated using the following formulas:

Discriminant: $(\Delta = b^2 - 4ac)$ **Roots:** $x_1 = \frac{-b + \sqrt{\Delta}}{2a} and\ x_2 = \frac{-b - \sqrt{\Delta}}{2a}$

Write a Python program that takes the coefficients a, b, and c as arguments, calculates the **real roots** of the equation, and prints them to the screen.

- Add the following logic based on the value of the discriminant:
  - **If** $(\boldsymbol{\Delta > 0})$**:** Print *"There are two distinct real roots:"* along with both roots.
  - **If** $(\boldsymbol{\Delta = 0})$**:** Print *"There is one repeated real root:"* along with the root $((x_1 = x_2 = -\frac{b}{2a})$.
  - **If** $(\boldsymbol{\Delta < 0})$**:** Print *"There are no real roots (the roots are complex)."*
  - *Print Sum of the roots:* $(x_1 + x_2 = -\frac{b}{a})$ *and Product of the roots:* $(x_1 * x_2 = \frac{c}{a})$